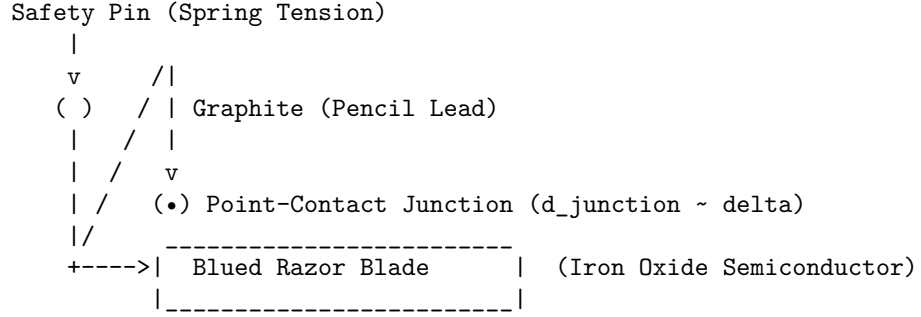


Within the flat 6D Conformal Spacetime Algebra system ($Cl(4, 1, 1)$), an amplitude modulation (AM) detector constructed from a safety pin, graphite pencil lead, and a blue razor blade (classically known as a *foxhole radio* detector) can be modeled as a point-contact Schottky diode.

This model describes how the mechanical junction rectifies ballistic wavefront trains [5.5, 5.8] and extracts the low-frequency envelope using localized soliton transitions and passive filtering.

Section 5.9: The Foxhole Radio Detector (Point-Contact Oxide Rectifier)

This detector consists of a spring-loaded metal safety pin holding a graphite (semimetal) tip in light point-contact with a “blue” steel razor blade. The “blueing” process creates a thin, chemically grown layer of iron oxide (magnetite, Fe_3O_4), which functions as an n-type semiconductor.



5.9.1 The Point-Contact Potential Barrier Because the contact area between the graphite tip and the thin iron oxide layer is extremely small, it forms a microscopic junction. The depletion layer width d_{junction} is comparable to the contact’s close-range mechanical softening parameter [Section 3]:

$$d_{\text{junction}} \sim \delta_{\text{contact}}$$

The work function difference between the graphite semimetal and the Fe_3O_4 oxide semiconductor creates an asymmetric Schottky potential barrier $V_{\text{barrier}}(x)$ across this narrow junction.

5.9.2 Wavefront Rectification An incoming amplitude-modulated electromagnetic wave $F_{\text{wave}}(t)$ [5.5], propagating ballistically as a series of convective wavefronts [5.8], is intercepted by an external antenna coil. This induces a high-frequency alternating potential $V_{\text{RF}}(t)$ across the point-contact junction:

$$V_{\text{RF}}(t) = V_c [1 + m(t)] \cos(\omega_{\text{carrier}} t)$$

where: * $m(t)$ is the low-frequency audio modulation signal. * ω_{carrier} is the high-frequency carrier wave.

This potential shifts the internal frequencies of the mobile electron solitons (ω_{metal} in the graphite) relative to the localized valence states (ω_{oxide} in the iron oxide lattice) [5.6]:

$$\hbar\omega_{\text{metal}}(t) = \hbar\omega_{\text{oxide}} + eV_{\text{RF}}(t)$$

Due to the asymmetric profile of the Schottky barrier, the transition rate Γ_{trans} of electron solitons [4.1] across the boundary is highly sensitive to the sign of V_{RF} :

1. **Forward Bias** ($V_{\text{RF}} > 0$): The barrier width is compressed ($d_{\text{junction}} < \delta_{\text{contact}}$). The field envelopes of the electron loops in the graphite overlap with vacant states in the oxide semiconductor, resulting in a high transition rate and charge flow.
2. **Reverse Bias** ($V_{\text{RF}} < 0$): The barrier width expands beyond the electron soliton's outer directrix radius ($d_{\text{junction}} \gg r_d$). This prevents field envelope overlap, shutting off the transition flow ($\Gamma_{\text{trans}} \approx 0$).

The asymmetric junction rectifies the current of electron solitons, governed by the non-linear diode relation:

$$I_{\text{rect}}(t) = I_0 \left(e^{\frac{eV_{\text{RF}}(t)}{k_B T}} - 1 \right)$$

This rectification converts the zero-mean alternating potential $V_{\text{RF}}(t)$ into a unidirectional, high-frequency pulsed current $I_{\text{rect}}(t)$ whose local average tracks the low-frequency modulation envelope.

5.9.3 Low-Pass Filtration and Envelope Extraction The rectified current $I_{\text{rect}}(t)$ is routed to a high-impedance telephone earphone. The receiving system forms a passive low-pass filter due to its inherent components: * The point-contact junction possesses a parasitic capacitance C_{junction} in parallel with the semiconductor boundary. * The earphone consists of a coil with high inductance L_{phone} and series resistance R_{phone} .

The high-frequency carrier component of the soliton current is bypassed through the low-impedance path of the junction capacitance C_{junction} .

However, because the audio modulation frequency is far lower than the carrier frequency ($\omega_{\text{audio}} \ll \omega_{\text{carrier}}$), the earphone coil presents low inductive reactance to the modulation envelope. The low-frequency current component $I_{\text{audio}}(t)$ flows directly through the earphone coil:

$$I_{\text{audio}}(t) \propto m(t)$$

This low-frequency current drives a varying magnetic field in the earphone coil, which exerts a fluctuating force on the earphone's metallic diaphragm. The diaphragm vibrates mechanically, releasing acoustic wavefronts that correspond to the original audio signal $m(t)$, completing the AM detection process.